

ANTIBACTERIAL DRUGS

(Cell wall and Protein synthesis inhibitors)

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INHIBITORS OF BACTERIAL CELL WALL SYNTHESIS

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OVERVIEW

- ▶ The bacterial cell wall is very important and performs the following functional roles;
 - I. Maintaining the shape of the bacterium
 - II. Protecting the bacterium from osmotic lysis in case it is found in an hypotonic media
- ▶ The bacteria cell wall primarily consists of **peptidoglycan**, a polysaccharide which is formed by the repeating disaccharide units namely the **N-acetylglucosamine** and **N-acetyl muramic** acid which are attached to each other through the **glycosidic** bonds
- ▶ The strands of peptidoglycan in the cell wall are cross linked by a transpeptidase reaction where the glycine pentapeptide on one strand is attached the penultimate D-alanine molecule on another end
- ▶ Without a cell wall, the bacterium is unprotected and this vulnerability leads to death

Mechanism of action

- ▶ A number of drugs work against cell wall synthesis whose effect is usually bactericidal
- ▶ This is actualized by preventing the crosslinking or elongation of peptidoglycan while on the other hand leading to bacterial autolysis
- ▶ It is worthy noting that cell wall synthesis take place during bacterial replication and therefore drugs that inhibit cell wall synthesis are more active against rapidly dividing bacteria than they are against resting or stationary phase of bacteria
- ▶ For the same reason, the effectiveness of cell wall inhibitors is sometimes reduced by concurrent administration of bacteriostatic antibiotics that slow the growth of bacteria

Examples on inhibitors

► The following group of drugs are inhibitors of cell wall synthesis;

Beta lactam drugs

- I. Penicillins
- II. Cephalosporins
- III. Carbapenems
- IV. Monobactams

Others

- I. Bacitracin
- II. Fosfomycin
- III. Vancomycin

PENICILLINS

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- ▶ Penicillins were the first antibiotic to have been isolated from microorganisms by Alexander Fleming which is used for treatment of bacterial infections
- ▶ Made up of a beta lactam ring consisting of a thiazolidine ring to which a unique chemical structure (R group) is attached and can be substituted with a different group in order to produce semi synthetic penicillins
- ▶ The route of administration is dependent on the stability of the drugs in gastric acid, with Acid stable penicillins being effective when given orally while acid labile ones must administered parenterally
- ▶ Penicillins are widely distributed to organs distributed except the central nervous system and because they penetrate the cerebrospinal fluid when the meninges are inflamed, they can be prescribed for treatment of bacterial meningitis
- ▶ Probenecid competes with penicillins for organic acid transporter in the proximal tubule with the former mostly slowing down the excretion of the later, the effect which has found a clinical use of prolonging the half life of penicillins

Classification of penicillins

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Penicillins can be grouped according to their antimicrobial activity and also partially based on their pharmacokinetic properties

1. Narrow spectrum (Natural penicillins)

- ▶ Penicillin G (Benzyl penicillin)- Intermediate acid resistant and can be given orally in larger doses
- ▶ Phenoxymethyl penicillin (Pen V) also referred to as acid resistant penicillin as it is able to resist GIT secretions hence administered orally

2. Penicillinase resistant penicillins (Beta lactamase resistant)

- ▶ Methicillin
- ▶ Nafcillin
- ▶ Dicloxacillin
- ▶ Flucloxacillin
- ▶ Cloxacillin

3. Extended spectrum penicillins (Broad spectrum penicillins)

3.1 Amino-penicillins

- ▶ Ampicillin
- ▶ Amoxycillin

3.2 Antipseudomonal penicillins

- ▶ Piperacillin
- ▶ Ticarcillin

4. Long Acting penicillins

- ▶ Procaine penicillin
- ▶ Benzathine penicillin

5. Beta Lactamase inhibitors

- ▶ B-lactamases are enzymes which inactivate penicillins and renders them ineffective and this process has significantly contributed to the emergence of microbial resistance
- ▶ There are currently four major classes of B-lactamase enzymes i.e. class A, B, C and D
- ▶ B-lactamases inhibitors are drugs which inhibit the molecular class A β -betactamases and thus prevent the inactivation of penicillins
- ▶ These drugs have no significant antimicrobial activity on their own but act as **surrogate substrates** also referred to as **suicide inhibitors**
- ▶ Examples of b-lactamase inhibitors are **Clavulanate (Clavulanic acid), Sulbactam, tazobactam**
- ▶ Combined examples include; **amoxicillin plus clavulanate, ampicillin plus sulbactam, piperacillin plus tazobactam, ticarcillin plus clavulanate**

Indications for Penicillins;

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Penicillin G

- ▶ Streptococcal infections

Penicillinase resistant penicillins

- ▶ Staphylococcal infections in penicillinase expressing bacteria
- ▶ Serious staphylococcal infections such as acute endocarditis, osteomyelitis as well as skin and soft tissue infections
- ▶ Staphylococci that are resistant to penicillinase resistant penicillins are referred to as methicillin resistant staphylococcus aureus (MRSA), methicillin having been the first drug in this group (Bacteria that are resistant to methicillin are also cross resistant to other drugs in the group)

Amino penicillins (Amoxicillin and ampicillin)

- ▶ Streptococcal and some strains of enterococci infections and a limited gram negative bacilli
- ▶ Amoxicillin is used alone for respiratory tract infections caused sensitive bacteria, otitis media, sinusitis, bronchitis, urinary tract infections and community acquired pneumonia

Indications for penicillins Cont'd

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Amoxicillin + clavulanate combination

- ▶ *Haemophilus influenzae* and *Moraxella catarrhalis*, *Streptococcus pneumoniae*
- ▶ Bite wound infection (infection mostly caused by *Pasteurella* & *S. aureus*)

Piperacillin (Anti-pseudomonal penicillin)

- ▶ Active against broad spectrum of gram-positive and gram-negative aerobic and anaerobic bacteria

Piperacillin + tazobactam

- ▶ Effective in patients with intra-abdominal, skin and soft tissue, lower respiratory infections, complicated UTI, gynaecological infections, febrile neutropaenia

Adverse effects

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- ▶ Except hypersensitivity reactions, penicillins are remarkably non toxic to the human body and produce very few adverse effects
- ▶ High concentrations may cause seizures due to irritation on the CNS
- ▶ Disturbance on the normal flora leading to diarrhea and superinfections with penicillin resistant organisms such as *Staphylococcus aureus* and *Clostridium difficile*
- ▶ *Pseudomonas colitis* can occur in association with *Clostridium difficile* superinfections

CEPHALOSPORINS

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- ▶ Cephalosporins are one of the most widely used group of antibiotics and they have a beta lactam ring and a dihydrothiazine ring
- ▶ Unlike penicillins, cephalosporins have at least two R groups attached to the molecule and hence enabling the synthesis of greater number of derivatives with potentially useful properties group and so the mechanism of action is as described with penicillins
- ▶ By manipulating the structure of cephalosporins, it has been possible to obtain drugs with greater resistance to bacterial Beta lactamases and with a wider range of antimicrobial activity
- ▶ When compared to penicillins, cephalosporins are more stable in the body and less likely to form antigens that evoke hypersensitive reactions
- ▶ The route of administration depends on the particular drug being used with most being either for oral or parenteral alone while cefuroxime is one of the few available cephalosporins that can be administered both orally and parenterally

Classification of cephalosporins

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Classification of cephalosporins is done based on the differences in their antimicrobial spectrum and they have been divided into five generations;

1. First generation

- ▶ Cephalexine
- ▶ Cefradine
- ▶ Cefadroxil

2. Second generation

- ▶ Cefaclor
- ▶ Cefuroxime
- ▶ Cefprozil
- ▶ Cefotetan
- ▶ Cefoxitin

3. Third generation

- ▶ Cefotaxime
- ▶ Cefotriaxone
- ▶ Cefpodoxime
- ▶ Ceftazidime

4. Fourth generation

- ▶ Cefepime

5. Fifth generation

- ▶ Ceftaroline
- ▶ Generally, the first generation drugs are primarily active against gram positive cocci and a limited gram negative bacilli
- ▶ The higher generation drugs have increased activity against gram negative bacilli and are less active against some species of gram negative cocci

Indications for cephalosporins

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First generation drugs

- ▶ Good activity against most streptococci and methicillin sensitive staphylococci
- ▶ Few gram negative bacilli like *Escherichia coli*, *Klebsiella pneumoniae*
- ▶ Treatment of Skin and soft tissue infections and uncomplicated UTI

Second line drugs

- ▶ Demonstrate similar with first generation drugs but while demonstrating activity against gram negative bacilli e.g. *H.influenza* in treating RTI, otitis media especially when proved to be resistant against amoxyl, UTI

Third generation

- ▶ Active against wide range of gram negative bacteria including *H.influenzae*, *M.catarrhalis*
- ▶ In addition ceftazidime is active against some strains of *P.aeruginosa*
- ▶ Gonococcal infections where they are mostly used as single doses in gonorrhea

Fourth generation drugs

- ▶ Cefepime is the example of the fourth generation cephalosporins
- ▶ It is active against a wide range of gram negative bacilli and is useful in severe infections where the causative organisms would have developed resistance to the other drugs

MONOBACTAMS

Aztreonam

- ▶ As a monocyclic B-lactam antibiotic, aztreonam is active against many aerobic gram negative bacilli like Enterobacter, Citrobacter, klebsiella, Proteus and P. aeruginosa
- ▶ Most useful in treatment of serious infections of susceptible organisms especially the multi drug resistant strains
- ▶ Aztreonam can cause hypersensitivity and thrombophlebitis
- ▶ It rarely shows cross sensitivity with penicillins and cephalosporins and so can be used in persons allergic to other B-lactam antibiotics

CARBAPENEMS

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- ▶ These are penicillin like drugs in which the Sulphur atom of the thiazolidine ring is replaced with a carbon atom and examples are **Imipenem, Doripenem, Ertapenem, Meropenem**
- ▶ These agents are bactericidal to many gram positive and gram negative bacteria including many aerobic and anaerobic gram negative bacilli and are resistant to many B-lactamases
- ▶ Carbapenems are used to treat a wide range of systemic infections including endocarditis, pneumonia, UTIs, pelvic, skin and soft tissue and intra-abdominal infections
- ▶ While the other drugs are free and safe, Imipenem is inactivated by renal dihydropeptidase enzyme and thus it is mostly combined with **Cilastatin** which is a dihydropeptidase inhibitor
- ▶ Carbapenems exhibit cross sensitivity with penicillins and other B-lactam antibiotics and so should not be administered to patient who are allergic to these drugs
- ▶ Though well tolerated, carbapenems can cause seizures in patients with epilepsy and less commonly anemia, leukopenia, thrombocytopenia and altered bleeding time

OTHER BACTERIAL CELL WALL SYNTHESIS INHIBITORS

Glycopeptides

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- ▶ **Vancomycin** and **Telavancin** its synthetic derivative are the examples of glycopeptides, the drugs which are active against gram positive cocci and gram positive bacilli and also proved to be active against MRSA and is usually is the first drug of choice for treating skin and soft tissue infections caused by these organisms
- ▶ Vancomycin is also used to treat streptococcal and enterococcal infections caused by penicillin resistant organisms like endocarditis and necrotizing fasciitis
- ▶ However some staphylococci and enterococci have developed some resistance through mutations that alters the amino acid sequence of the cell wall pentapeptide containing D-alanine
- ▶ Although proved to be effective against diarrhea and pseudomonas colitis caused by C. difficile, metronidazole is usually preferred
- ▶ Vancomycin can cause ototoxicity and nephrotoxicity and should be used cautiously with other nephrotoxic drugs like amphotericin B and aminoglycosides (Improved formulations have reduced incidence)
- ▶ Red neck or red man syndrome is an erythematous rash that develops on the face the upper body when vancomycin is infused at an excessive rate

Bacitracin

- ▶ This drug inhibits synthesis of the cell wall peptidoglycan by blocking the regeneration of bactoprenol phosphate, the lipid carrier molecule
- ▶ It is very nephrotoxic and thus it is not administered systemically but mostly used in combination with neomycin and polymyxin for the topical treatment of minor skin and ocular, it being active against staphylococci and streptococci

Fosfomycin

- ▶ This drug is unique in that it blocks formation of portion of N-acetylmuramic acid (UDP-MurNAc) which is one of the first steps in cell wall peptidoglycan synthesis
- ▶ The drug is active against enterococci and many gram negative enteric bacilli including *E.coli*, *Klebsiella species*, *Citrobacter*, *Proteus species*
- ▶ It is recommended for treatment of uncomplicated UTIs caused by *E.coli* or *E.faecalis*
- ▶ Fosmycin sometimes causes diarrhea but is otherwise well tolerated and is associated with few adverse effects

INHIBITORS OF BACTERIAL PROTEIN SYNTHESIS

OVERVIEW

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- ▶ Bacteria are prokaryotes and their protein synthesis is different from that of mammals and other eukaryotes
- ▶ Each ribosome contains two subunits, with the prokaryotes having 30s and 50s subunit while the eukaryotes have 40s and 60s subunit
- ▶ The basic step in bacterial protein synthesis include the binding of aminoacyl transfer RNA (tRNA) to the ribosome, the formation of peptide bond and translocation
- ▶ Aminoacyl tRNA binds to 30s ribosomal subunits while the peptide bond and translocation involve the components of the 50s ribosomal subunit
- ▶ Several classes of antibiotics act by selectively blocking one or more steps in the protein synthesis of bacteria
- ▶ This structural and functional difference in prokaryotic and eukaryotic cells is the basis on which the drug selectivity for bacterial protein synthesis is anchored

Classes of drugs that inhibit protein synthesis

The following drugs inhibit protein synthesis by either targeting the 30s subunits or the 50s subunit as sites of drug action as shown below;

Drugs that target the 30s subunit

- ▶ Tetracycline
- ▶ Aminoglycoside

Drugs that target 50s subunit

- ▶ Macrolides
- ▶ Chloramphenicol
- ▶ Clindamycin
- ▶ Dalbavipristin

DRUGS TARGETING 30s RIBOSOMAL SUBUNIT

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1. AMINOGLYCOSIDES

- ▶ Examples include **amikacin, gentamycin, kanamycin, neomycin, tobramycin**
- ▶ Aminoglycosides consists of amino sugars linked through glycosidic bonds, these amino sugars being highly basic and being subject to extensive protonation and ionization in body fluids
- ▶ This leads to poor absorption from the gut and hence aminoglycosides must be administered parenterally for treatment of systemic infections
- ▶ They are occasionally administered orally for GIT infections e.g. neonatal necrotizing enterocolitis while topical preparations are used to treat infections of the skin, mucous membrane and ocular tissues
- ▶ They are poorly absorbed in body tissues due to their highly ionized nature and cannot cross the meninges whether inflamed or not and so intrathecal injections may be preferred for meningitis treatment
- ▶ The clearance of aminoglycosides is equal to creatinine clearance and being proportional to glomerular filtration rate, the dosage of aminoglycosides must be reduced in patients with renal impairment

Indications

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- ▶ Aminoglycosides are active against a wide range of aerobic gram negative bacilli

Streptomycin

- ▶ The least active against most gram negative bacilli and is used to treat T.B, infections caused by *Yersinia* (Plague), *Fransisella tularensis* (tularemia)

Tobramycin

- ▶ The most active aminoglycoside against many strains of *Pseudomonas aeruginosa*

Gentamycin

- ▶ More active against *E.coli*, *Klebsiella* species
- ▶ Used in combination with a penicillin to treat enterococcal, staphylococcal or viridans group like streptococcal infections like endocarditis

Amikacin

- ▶ Used to treat infections caused by strains resistant to gentamycin and tobramycin as it is more resistant against bacteria enzymes

Adverse effects

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The most serious adverse effects are nephrotoxicity and ototoxicity

- ▶ Risk of toxicity is related to the dosage and duration of treatment and varies with specific drug
- ▶ Irreversible toxicity can occur even after use of the drug is discontinued however, serious toxicity is less likely when the offending drug is discontinued at an earliest sign of dysfunction
- ▶ Aminoglycosides are known to be some of the most causes of drug induced renal failure
- ▶ Ototoxicity is associated to accumulation of aminoglycosides in the labyrinth and hair cells of the cochlea
- ▶ The aminoglycosides vary in their tendency to cause cochlea toxicity i.e. Amikacin and tobramycin cause more cochlea toxicity (Deafness), Neomycin is the most nephrotoxic and hence its use is limited to topical treatment of superficial infections

2. TETRACYCLINES

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- ▶ Tetracyclines are four ring anthracycline compounds produced by *Streptomyces* species
- ▶ Examples are doxycycline, minocycline, tetracycline, tigecycline
- ▶ Oral bioavailability of the tetracycline varies from 70% of tetracycline to over 90% of doxycycline and minocycline
- ▶ All tetracycline bind to divalent and trivalent cations like calcium, aluminium and iron and hence dairy products, anti-acids and other drug combinations or foods containing these elements should not be taken at the same time with tetracyclines as they reduce the later's bioavailability
- ▶ Unlike other tetracyclines, doxycycline does not depend on renal elimination and so there is no need to adjust the doses in patients with renal insufficiency

Indications

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- ▶ Tetracycline are broad spectrum bacteriostatic drugs that inhibit growth of many gram positive and gram negative organisms
- ▶ Infections caused by rickettsia and also lyme disease and other diseases caused by borrelia burgdorferi
- ▶ Infections caused by spirochaetes like treponema pallidum (syphilis) and other genital infections caused by chlamydia trachomatis like PID
- ▶ Tetracyclines are alternatives to macrolides to treat infections caused by mycoplasma species
- ▶ Acne vulgaris by suppressing the growth of *Propionibacterium acnes*, an organism that converts sebum triglycerides to fatty acids which irritate the skin, cause inflammation of the sebaceous gland and comedone formation
- ▶ *Vibrio cholera* where they shorten the disease course of cholera and also help prevent the transmission of the disease to others
- ▶ Peptic ulcers caused by *H.pylori*

Adverse effects

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- ▶ Discoloration of teeth due to their tendency to accumulate in bone tissue and hence they should not be used in pregnant women or children below the age of 8
- ▶ Potentially severe nephrotoxicity and hepatotoxicity in form of fatty degeneration though rare but makes it another reason why pregnant women should not take them as they are more prone
- ▶ Degraded tetracycline have been found to be more nephrotoxic than the parent drug and hence it is advisable to discard must not be used beyond their expiration date
- ▶ Photosensitivity in individuals who are exposed to the sun during therapy, doxycycline being more of a culprit than tetracycline and minocycline

DRUGS THAT TARGET 50s RIBOSOMAL SUB UNITS

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1. MACROLIDES

- ▶ Examples include **erythromycin**, **azithromycin** and **clarithromycin**
- ▶ Clarithromycin and azithromycin are semisynthetic derivatives of erythromycin which have improved pharmacokinetic properties and antibacterial activity
- ▶ Macrolides are usually administered orally though azithromycin and clarithromycin are available in intravenous formulation for treatment of serious infections like legionnaire disease
- ▶ Oral bioavailability of erythromycin is lower than that of clarithromycin and azithromycin which achieve higher tissue concentrations

Indications

- ▶ Macrolides are active against many gram positive and gram negative that cause upper respiratory tract infections and pneumonia
- ▶ The antimicrobial coverage includes *Streptococci*, *pneumococci*, *chlamydiae*, *M. pneumonia* and *Legionella pneumophila*
- ▶ Azithromycin is active against pathogens responsible for sinusitis, otitis media and bronchitis
- ▶ Active against *Chlamydia pneumoniae* and *Chlamydia trachomatis* and hence is used for pneumonia and genitourinary infections caused by respective organisms
- ▶ Azithromycin and clarithromycin can be used to treat *Mycobacterium avium* infections occurring in AIDS
- ▶ Clarithromycin is the most active macrolide against *H.pylori* and thus it is used in treatment of peptic ulcers

Adverse effects

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- ▶ Commonly causes stomatitis, heartburn, nausea, anorexia, abdominal discomfort, diarrhea
- ▶ Erythromycin binds to receptors for motilin, a gastric hormone that activates duodenal and jejunal receptors initiate peristalsis
- ▶ Activation of these receptors by erythromycin causes uncoordinated peristalsis leading to anorexia, nausea and vomiting
- ▶ Azithromycin and clarithromycin have less affinity for motilin receptors and hence cause less GIT distress than erythromycin
- ▶ Large i.v doses of erythromycin cause ototoxicity in the form of tinnitus or impaired hearing which mostly subside when the drug is discontinued

Drug interactions

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- ▶ Erythromycin and clarithromycin inhibit cytochrome P450 isozyme 3A4 (CYP3A4) and thus can elevate the plasma concentration of a large number of drugs metabolized by this isozyme
- ▶ Concurrent administration of erythromycin or clarithromycin with carbamazepine can lead to life threatening carbamazepine toxicity and this combination should be avoided
- ▶ Clarithromycin and erythromycin also inhibit metabolism of statins like simvastatin and lovastatin leading to elevated statin levels and rhabdomyolysis
- ▶ In contrast, azithromycin has little effect on CYP450 making it a drug of choice for those taking other drugs

2. CLINDAMYCIN

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- ▶ Clindamycin is a chlorinated derivative of lincomycin antibiotic that was isolated from *Streptomyces* species
- ▶ Lincomycin is less active than clindamycin and hence it is no longer used

Indications

- ▶ Clindamycin is active against gram positive cocci and anaerobic organisms such as *B.fragilis* and *Clostridium perfringens* (cause of gas gangrene)
- ▶ Treatment of infections caused by MRSA including necrotizing fasciitis

Adverse effects

- ▶ GIT superinfections caused by *Clostridium difficile*
- ▶ These superinfections lead to diarrhea and Pseudomembranous colitis and hence patients who develop diarrhea when on clindamycin should stop discontinue therapy

3. Chloramphenicol

- ▶ Chloramphenicol is highly lipophilic and hence it is well absorbed from the GIT leading to high concentrations in the CNS even in the absence of inflamed meninges
- ▶ Chloramphenicol can either be bacteriostatic or bactericidal depending on the organism and the drug concentration
- ▶ This property makes chloramphenicol an effective treatment of meningitis caused by sensitive organisms
- ▶ Chloramphenicol is metabolized partly by glucuronate conjugation with the parent and metabolite being excreted in urine
- ▶ Neonates have reduced ability to conjugate the drug and hence if doses are not reduced in neonates, the drug accumulates in the plasma and causes gray baby syndrome (Ashen gray cyanosis, weakness, respiratory depression, hypotension and shock)

Indications

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- ▶ Chloramphenicol is broad spectrum antibiotic which is active against Pneumococci, meningococci and H.influenza which are the major causative organisms of meningitis
- ▶ Has also been used to treat salmonella and Bacteroides infections
- ▶ Generally, chloramphenicol is usually reserved for treating meningitis and other infections caused by organisms that are resistant to other drugs and for infections in persons who are allergic to less toxic antibiotics

Adverse effects

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- ▶ **Gray baby syndrome** (Ashen gray cyanosis, weakness, respiratory depression, hypotension and shock)
- ▶ This occurs because neonates have reduced ability to conjugate the drug and hence if doses are not reduced in neonates, the drug accumulates in the plasma
- ▶ The other adverse effects in bone marrow suppression leading to potentially fatal aplastic anaemia or dose dependent anaemia

OTHER PROTEIN SYNTHESIS INHIBITORS

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1. LINEZOLOID

- ▶ Binds to 23S RNA component of the 50S ribosomal sub unit and prevent the 70S initiation complex required for bacterial protein synthesis
- ▶ Because of its unique mechanism, cross resistance with other classes of antibiotics is unlikely while Its bioavailability is almost 100%

Indications

- ▶ Treatment of infections caused by vancomycin resistant E.faecium, pneumonia caused by MRSA and skin and soft tissue infections methicillin sensitive or MRSA, streptococcus pyogenes, MDR TB

Adverse effects

- ▶ Causes thrombocytopaenia in patients with renal insufficiency or during prolonged therapy

2. MUPIROCIN

- ▶ An antibiotic obtained from *Pseudomonas fluorescens*
- ▶ Active against most staphylococci including many methicillin resistant strains
- ▶ Also inhibits most beta haemolytic streptococci including *S.pyogenes*
- ▶ Mupirocin is the first effective topical therapy against impetigo, a skin disease caused by streptococci and staphylococci
- ▶ Mupirocin is also used to eradicate nasal colonization of MRSA in infected patients and in health workers

END